NAVIER-STOKES BRIEF

Can a smooth fluid become infinitely rough?

The 3D Navier–Stokes equations describe water and air. The Millennium Problem asks whether every smooth, physically reasonable starting flow stays smooth for all time—or whether a singularity can form in finite time.



THE QUESTION IN ONE LINE

Smooth forever or **finite blow-up?**

No generally accepted proof decides this in three dimensions.

Our repository tests proof routes with C++ and Lean.

It has not solved the Millennium Problem.

Conceptual illustration — not a simulation result

Stretching creates small scales; viscosity removes them

The unresolved issue is whether the smoothing always wins before any derivative becomes unbounded.

$$\partial u/\partial t + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \qquad \nabla \cdot u = 0$$

velocity changes = transport + pressure + viscous smoothing; incompressibility forbids sources or sinks

VORTEX STRETCHING

Can sharpen gradients

- Three-dimensional vortices can stretch and intensify.
- Energy may move toward smaller spatial scales.
- A proof must rule out uncontrolled concentration.

NONLINEAR TERM $(u \cdot \nabla)u$

UNKNOWN BALANCE

VISCOSITY

Smooths the flow

- Diffusion damps high-frequency motion.
- Energy decreases in the classical energy estimate.
- Known estimates are not yet strong enough to close 3D regularity.

DIFFUSIVE TERM $\nu \Delta u$

A successful proof must control every scale uniformly—not only every scale tested by a computer.

NOT SOLVED

Lean has formally rejected five proposed unrestricted PNT bounds

A six-mode Fourier witness makes PNT-2/7/8/12/14 unbounded as its amplitude tends to zero. The continuous Clay problem remains open.

LAYER	WHAT THE PROJECT DOES	CURRENT RESULT	STATUS
C++ spectral lab	Search finite Fourier states and compare outputs	Numerical consistency checks; not a proof premise	VERIFIED
Lean witness	Exact six-mode and full-cube calculations	Unrestricted PNT-2/7/8/12/14 bounds refuted	PROVED
Lean identities	Cauchy chain and finite Gram-Schur estimate	75 theorems; 259 declarations audited	PROVED
PNT-13	Standalone shell decorrelation	Strong finite counterevidence; no theorem	OPEN
Continuous PDE	Local closure and L4–L6 regularity chain	No solution of the Millennium Problem	OPEN

HOW TO VERIFY THE LEAN RESULT

Run ./navier-stokes/lean/verify.sh. It checks the proof library, audits axioms and rejects four deliberate bad proofs.

WHAT REMAINS

Reformulate PNT, certify the local sign candidate, and close the continuous L4–L6 argument.